QUIZ 6

Section B

Q1) For every integer n >= 8, n cents can be obtained using 3 cents and 5 cents coins.

Proof (by mathematical induction):
Let the property $P(n)$ be the sentence

n¢ can be obtained using 3¢ and 5¢ coins. ← $P(n)$

Show that P(8) is true:

$P(8)$ is true because 8¢ can be obtained using one 3¢ coin and one 5¢ coin.

Show that for every integer $k \geq 8$, if $P(k)$ is true then $P(k+1)$ is also true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 8$. That is:] Suppose that k is any integer with $k \geq 8$ such that

k¢ can be obtained using 3¢ and 5¢ coins. ← $P(k)$ inductive hypothesis

[We must show that $P(k+1)$ is true. That is:] We must show that

$(k+1)$¢ can be obtained using 3¢ and 5¢ coins. ← $P(k+1)$

Case 1 (There is a 5¢ coin among those used to make up the k¢.): In this case replace the 5¢ coin by two 3¢ coins; the result will be $(k+1)$¢.

Case 2 (There is not a 5¢ coin among those used to make up the k¢.): In this case, because $k \geq 8$, at least three 3¢ coins must have been used. So remove three 3¢ coins and replace them by two 5¢ coins; the result will be $(k+1)$¢.

Thus in either case $(k+1)$¢ can be obtained using 3¢ and 5¢ coins *[as was to be shown]*.

[Since we have proved the basis step and the inductive step, we conclude that the proposition is true.]

Q2) For any integer *n* >=1, if one square is removed from a 2^n x 2^n checkerboard, the remaining squares can be completely covered by L-shaped trominoes.

The main insight leading to a proof of this theorem is the observation that because $2^{k+1} = 2 \cdot 2^k$, when a $2^{k+1} \times 2^{k+1}$ board is split in half both vertically and horizontally, each half side will have length 2^k and so each resulting quadrant will be a $2^k \times 2^k$ checkerboard.

Proof (by mathematical induction):

Let the property $P(n)$ be the sentence

> If any square is removed from a $2^n \times 2^n$ checkerboard, then the remaining squares can be completely covered by L-shaped trominoes. ← $P(n)$

Show that $P(1)$ is true:

A $2^1 \times \times 2^1$ checkerboard just consists of four squares. If one square is removed, the remaining squares form an L, which can be covered by a single L-shaped tromino, as illustrated in the figure to the left. Hence $P(1)$ is true.

Show that for every integer $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is also true:

[Suppose that P(k) is true for a particular but arbitrarily chosen integer $k \geq 3$. That is:]
Let k be any integer such that $k \geq 1$, and suppose that

> If any square is removed from a $2^k \times 2^k$ checkerboard, then the remaining squares can be completely covered by L-shaped trominoes. ← $P(k)$

$P(k)$ is the inductive hypothesis.

[We must show that $P(k+1)$ is true. That is:] We must show that

> If any square is removed from a $2^{k+1} \times 2^{k+1}$ checkerboard, then the remaining squares can be completely covered by L-shaped trominoes. ← $P(k+1)$

Consider a $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed. Divide it into four equal quadrants: Each will consist of a $2^k \times 2^k$ checkerboard. In one of the quadrants, one square will have been removed, and so, by inductive hypothesis, all the remaining squares in this quadrant can be completely covered by L-shaped trominoes.

The other three quadrants meet at the center of the checkerboard, and the center of the checkerboard serves as a corner of a square from each of those quadrants. An L-shaped tromino can, therefore, be placed on those three central squares. This situation is illustrated in the figure to the left.

By inductive hypothesis, the remaining squares in each of the three quadrants can be completely covered by L-shaped trominoes. Thus every square in the $2^{k+1} \times 2^{k+1}$ checkerboard except the one that was removed can be completely covered by L-shaped trominoes *[as was to be shown]*.